

4.TRANSPORT VENTILATOR

TECHNICAL SPECIFICATION

EMERGENCY & TRANSPORT VENTILATOR

Intended Application:

For adult and pediatric patients requiring invasive or non-invasive mandatory and assisted ventilation during **intra-hospital transport, inter-hospital transfer, emergency care, ICU transport, ambulance and critical care transport.**

A. General Specifications

Sl. No.	Parameter	Minimum Tender Requirement
1	Type	Compact, robust, portable emergency and transport ventilator
2	Patient Category	Adult and Pediatric
3	Ventilation	Time-cycled, volume-controlled and pressure-controlled ventilation
4	Application	Suitable for invasive and non-invasive ventilation
5	Tidal Volume	50 mL to 2,000 mL or better
6	Ventilation Modes	Minimum VC-CMV, VC-AC, VC-SIMV, Spontaneous CPAP and PC-BIPAP or equivalent
7	Pressure Support	Available in applicable spontaneous and mandatory ventilation modes
8	NIV	Non-invasive ventilation capability
9	Apnoea Ventilation	Automatic apnoea ventilation facility
10	AutoFlow	Volume-targeted, pressure-controlled ventilation capability; integrated or available as an option
11	100% Oxygen	Dedicated 100% O ₂ function
12	O ₂ Inhalation	Oxygen inhalation facility, where applicable
13	Inspiration Hold	Available
14	Respiratory Rate	Minimum range 2–60 breaths/min , depending on ventilation mode
15	Inspiratory Time	0.2–10 seconds or better
16	I:E Ratio	Adjustable over a wide range, approximately 1:100 to 50:1 or better
17	Inspiratory Pressure	Minimum PEEP +3 to +55 mbar or better
18	PEEP/CPAP	0–20 mbar or better
19	Pressure Support	0–35 mbar or better , relative to PEEP
20	Flow Trigger	Adjustable, approximately 1–15 L/min or better

Sl. No.	Parameter	Minimum Tender Requirement
21	Maximum Inspiratory Flow	At least 100 L/min at appropriate supply pressure
22	Oxygen Concentration	Adjustable approximately 40–100% FiO₂
23	Slope/Rise Time	Adjustable pressure rise time with minimum slow, standard and fast settings
24	CPR Function	Pressure-limited ventilation behavior when maximum pressure is reached
25	Patient Interface	Compatible with endotracheal tube, tracheostomy and NIV masks

B. Monitoring & Display

Parameter	Minimum Tender Requirement
Display	Integrated graphical display suitable for transport use
Display Technology	High-contrast display readable in emergency/transport environments
Display Resolution	Minimum approximately 240 × 128 pixels or equivalent
Display Area	Minimum approximately 108 × 56 mm or equivalent
Waveforms	Real-time pressure and flow waveforms, as applicable
Measured Parameters	Tidal volume, minute volume, respiratory rate, FiO ₂ , PEEP, mean airway pressure and peak inspiratory pressure
Additional Monitoring	Plateau pressure and spontaneous minute volume, where applicable
EtCO ₂	Integrated mainstream capnography should be included
Capnography	Continuous EtCO ₂ monitoring with waveform display
Alarm Display	Visual and audible alarm indication
Alarm Parameters	High/low airway pressure, minute volume, respiratory rate, apnoea, FiO ₂ and technical/system alarms

C. Gas Supply

Parameter	Minimum Requirement
Gas	Medical-grade oxygen
Supply	Compatible with central pipeline and O ₂ cylinder
O ₂ Supply Pressure	270–600 kPa or equivalent
Gas Consumption	Low internal control gas consumption

Parameter	Minimum Requirement
Oxygen Connector	Suitable standard hospital/ambulance oxygen connector
Pressure Regulator	Compatible with approved medical oxygen pressure regulator

The published technical data specify medical oxygen supply from pipeline or cylinder with an O₂ supply pressure of **270–600 kPa**.

D. Battery & Power Supply

Parameter	Minimum Requirement
Internal Battery	Rechargeable internal battery
Battery Operation	Suitable for uninterrupted patient transport
External Power	AC/DC power supply capability
Ambulance Power	Compatible with vehicle DC power supply through suitable converter
Battery Indication	Remaining battery capacity/status displayed
Low Battery Alarm	Visual and audible indication
Charging	Battery charging during operation when connected to external power
Power Failure	Automatic changeover to internal battery
Battery Replacement	Battery shall be replaceable/serviceable by authorized personnel

Tender recommendation: The bidder should clearly state the **minimum guaranteed operating time on internal battery at specified test conditions** and provide a battery backup certificate. This is preferable to prescribing a model-specific value.

E. Physical Specifications

Parameter	Minimum Requirement
Design	Compact, rugged and suitable for emergency transport
Weight	Preferably ≤6 kg including internal battery
Dimensions	Preferably approximately 300 × 190 × 180 mm or compact equivalent
Mounting	Suitable for ambulance mounting and hospital transport trolley
Carrying	Integrated handle/carrying system
Protection	Robust housing suitable for emergency transport
Environmental	Suitable for hospital, ambulance and inter-hospital transport conditions
Noise	Low operating noise suitable for patient transport

F. Connectivity & Data

The ventilator should provide:

- Digital data interface for monitoring/data transfer.
- RS-232 or equivalent communication interface.
- MEDIBUS or equivalent communication protocol.
- Real-time data export capability.
- Compatibility with hospital/ambulance monitoring systems where applicable.

REQUIRED ACCESSORIES

The quoted system shall be supplied as a **complete operational package**, including:

1. Transport ventilator – **1 No.**
2. Rechargeable internal battery – **1 No.**
3. AC/DC power supply – **1 No.**
4. DC/DC vehicle power converter – **1 No.**
5. Oxygen high-pressure hose – **1 No.**
6. Patient breathing circuit – minimum **2 sets**
7. Reusable breathing circuit, if applicable
8. Adult reusable patient circuit – **1 set**
9. Pediatric patient circuit – **1 set**
10. NIV mask/interface – suitable range
11. Mainstream CO₂ sensor – **1 No.**
12. CO₂ cuvettes/adapters – initial supply
13. Oxygen cylinder connector/adaptor
14. Ambulance mounting bracket – **1 No.**
15. Transport trolley/carrying system, if required
16. Carrying bag/case – **1 No.**
17. User manual – **1 No.**
18. Service manual – **1 No.**
19. Complete installation and commissioning
20. On-site user training

SAFETY & ALARM REQUIREMENTS

The ventilator shall provide:

- High airway pressure alarm
- Low airway pressure alarm
- Apnoea alarm
- Low minute volume alarm
- High respiratory rate alarm
- Low/high FiO₂ alarm, where applicable
- Low oxygen supply pressure alarm

- Battery low alarm
- Power failure alarm
- Ventilator malfunction alarm
- Disconnection alarm
- Obstruction alarm
- Technical fault indication
- Audible and visual alarms
- Alarm silence/mute function with automatic reactivation

TENDER WARRANTY & SERVICE REQUIREMENTS

Parameter	Requirement
Warranty	Minimum 3 years comprehensive warranty from date of installation/commissioning
Battery Warranty	Minimum 2 years or as per manufacturer's standard, whichever is higher
Service	Authorized service support available in India
Response Time	Service engineer response within 24–48 hours
Preventive Maintenance	Minimum 2 preventive maintenance visits per year during warranty
Spare Parts	Availability for minimum 7–10 years
Training	Training for ICU/ER doctors, nurses, respiratory therapists and biomedical engineers
Installation	Installation, testing and commissioning included
Demonstration	Demonstration to technical committee if requested
Documentation	User manual, service manual, IFU and maintenance schedule

WARRANTY & AFTER SALES SUPPORT

- **Backup Machine:** In the event of any equipment complaint or breakdown, a suitable backup machine shall be arranged within 12 hours until the original equipment is restored to operational condition.
- **AMC/CAMC Charges:** The AMC/CAMC rate as a percentage of the equipment purchase value, applicable after the warranty period, shall be finalized and frozen at the time of purchase and clearly mentioned in the Purchase Order.
- **Manufacturer Authorization & Warranty:** All authorization letters, warranty documents, and related commitments shall be issued directly by the original manufacturer/company, and not solely by the vendor/dealer.

